



UL Campus Nav App

Project Report



Module Name: CS6431/41 – Foundations of Interaction and Experience Design

Assignment-4: Testing and Redesign

Project Title: UL Campus Nav App

Student Name: Christian Anene

Student ID: 25063456

Submission Date: 14 December 2025

Module Instructor: Dr. Cristiano Storni

Table of Content:

Sr no.	Content	Page no.
1	Introduction • 1.1 Overview • 1.2 Main Goal of the Redesign	3
2	Group Project Recap • 2.1 The Group Project Developed process • 2.2 The Group Project Developed features	5
3	Testing • 3.1 Type of Evaluation Selected • 3.2 Testing Method • 3.3 Participant Profile • 3.4 Key Findings • 3.5 Implications for Redesign	10
4	Exploring Design Direction • 4.1 Identifying Possible Directions • 4.2 Brainstorming and Mind Mapping	12
5	Designing • 5.1 Features • 5.2 Design Rationale • 5.3 Wireframes • 5.4 Final Prototype Screens	15

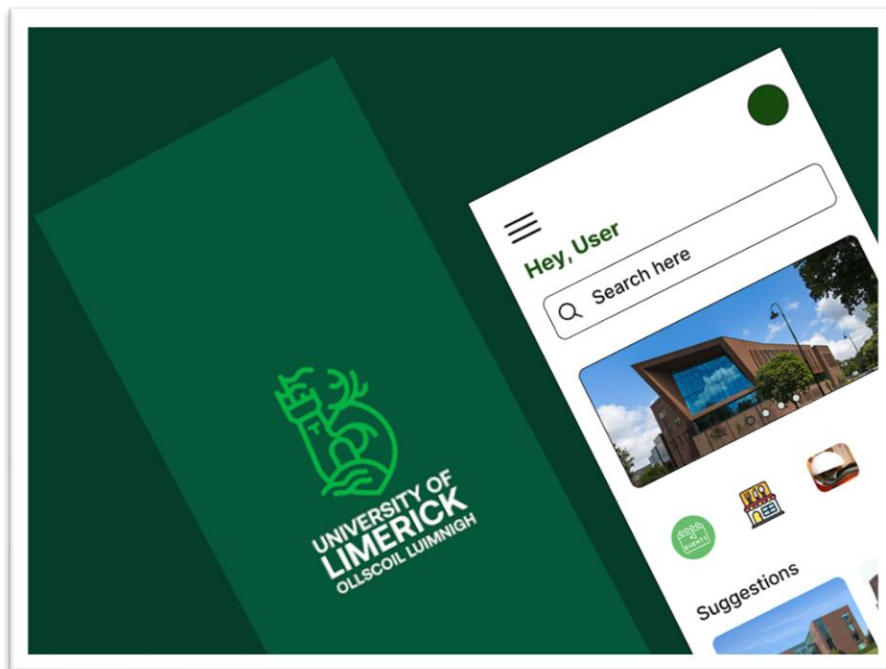
6	Evaluation of the Redesign • 6.1 Improved interface clarity • 6.2 Enhanced Feature Discoverability • 6.3 More Understanding Route Instruction • 6.4 More Predictable and Intuitive Navigation Flow • 6.5 Remaining Opportunity for Improvement	17
7	Conclusion	18
8	References	
9	Annexes • Annex I: User flow • Annex II: Prototype • Annex III: Prototype Connection	19

1.Introduction:

1:1 Overview:

Finding your way around a large university campus can be confusing and stressful. This is especially true for first year students, visitors, parents, and even staff who may not know where certain buildings or services are located. The University of Limerick (UL) is a good example of this challenge. The campus is large, split by the River Shannon, and filled with many academic buildings, cafés, sports areas, and student services. While this creates a lively and open environment, it also makes navigation difficult for many people.

The UL Campus Navigation App was initially developed as a group design intervention to address these wayfinding challenges. The group prototype introduced key features such as destination search, building information pages, interactive campus maps, and basic route instructions. The intention was to create a UL specific tool that reduces cognitive effort and provides clear guidance for locating buildings and understanding campus routes.



Base on the feedback after usability testing conducted during the group phase revealed both strengths and critical areas for improvement. Participants valued the concept of a dedicated UL navigation system but reported issues relating to interface clarity, feature discoverability, and Users found the app difficult to understand at first glance. The interface showed many buildings without clear explanations, so it wasn't obvious what each building was. The navigation bar layout was also not intuitive, its buttons and icons didn't feel logical, and users couldn't immediately tell what they were supposed to do. In addition, accessibility options were not clearly presented, making it hard for users to know which routes or entrances were close.

Taking this feedback into account, I created an improved personal redesign of the app. The new design focuses on simplifying the layout, making actions clearer, and improving the overall user flow. I reorganised the navigation into four easy to understand screens:

1. **Home / Search Page** – A simplified entry point with clear categories, a prominent search bar, and nearby building suggestions with simple explanation and name of the building.
2. **Place Detail Page** – A refined building page featuring a hero image, walking time, distance, accessibility indicators, and two clear actions: Start walking or Preview route.
3. **Route Preview Screen** – A lightweight waypoint view providing a mini-map, step-by-step instructions, and an option to begin navigation.
4. **Live Navigation Screen** – A full-screen guided map experience with a highlighted route, real-time progress, step bubbles, ETA, and a clearly distinguished End route button.

This approach aligns with Don Norman's concept of Design for Discoverability, ensuring that all essential navigation information remains visible, intuitive, and easily interpretable. Within the context of this module, the redesign makes full use of ideation,

sketching, prototyping, and iterative testing to evaluate alternative approaches and converge toward a clearer and more user-friendly solution.

The UL Campus Navigation redesign is therefore not simply a technical upgrade, but a deeper exploration of interaction and experience design principles. It seeks to evolve the app from a functional prototype into a thoughtful, user-centred navigation companion tailored to the realities of the UL university. This report documents that process what was redesigned, why it was necessary, and how the new prototype better supports wayfinding across the university campus. It follows the UCD (User-Centred Design) cycle and demonstrates the progression from conceptual exploration and collaborative prototyping to a refined, individual interactive prototype capable of guiding users through a complete navigation task.

1:2 Main Goal of Redesign and Objective

The main goal of this redesign is to improve the UL Campus Navigation App so that users can find buildings and navigate around the University of Limerick more easily, more confidently, and with less confusion. The redesign aims to create a clean, simple, and user-friendly experience that supports students, staff, and visitors as they move around the campus.

Specific Objectives:

1. Improve clarity and reduce confusion

- Simplify the layout and visual structure so users can easily understand.
- Reduce cognitive load by showing only the information needed at each step of the journey.

2. Make navigation faster and easier to follow

- Redesign the navigation flow into clear, logical stages: Search, Details, Route Preview, Live Navigation.
- Provide more understandable step-by-step directions with clear map visuals.

3. Address issues identified during usability testing

- Fix problems reported by users, such as confusing interface, and navigation bar issues.

-
- Integrate user feedback to create a smoother, more intuitive experience.

4. Improve accessibility for different user types

- Ensure that screens and icons are readable, well-spaced, and usable for people with varied needs.
- Provide clear visual and text-based guidance that works for both local and international users.

2.Group Project Recap:

The group phase of the UL campus Nav App project focused on examining everyday challenges encountered within the UL campus environment like navigating large and unfamiliar areas, locating facilities, and accessing real time information about the campus services. We used creative exploration methods where the group was able to differ initially by generating numerous ideas and then converge on a feasible and concept of the UL Campus Nav App.

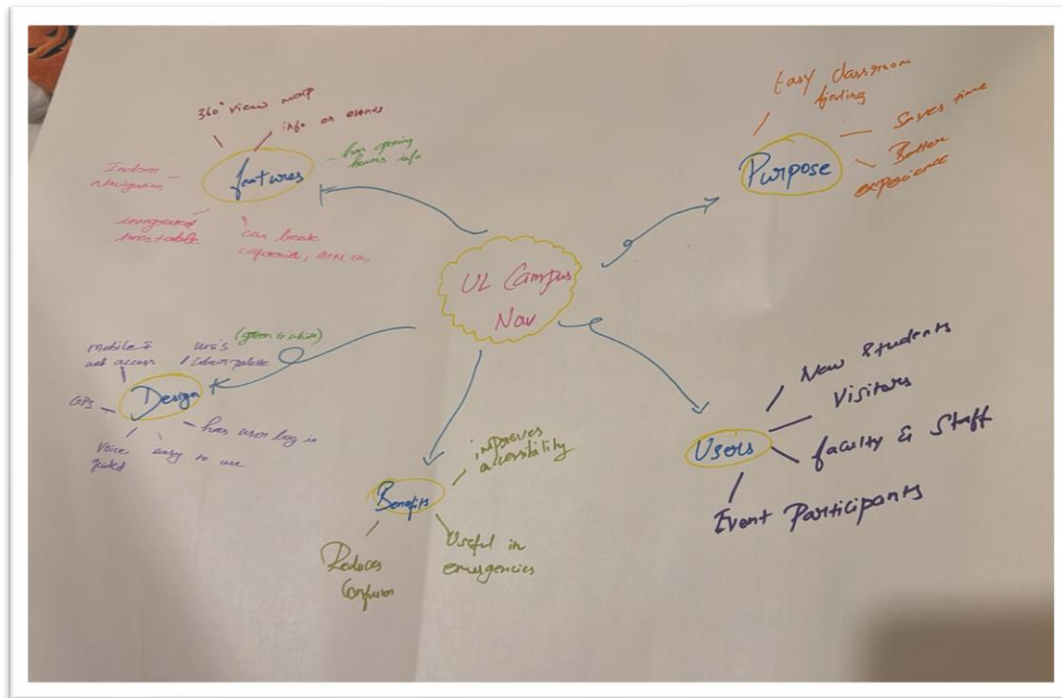
The aim of these actions was to identify opportunities to involve an interactive design which will enhance the on-campus experience for students, staff and visitors at University of Limerick.

2.1 The Group Project Developed Process:

Brainstorming technique: from the perspective of group discussion, the brainstorming technique was adopted during the process. Indeed, while the team first started presenting their ideas during the campus group, all the ideas were accepted and written down by the moderator. In other words, the group continued an informal conversation and was not specifically focused on the UL Nav App topic.

Although each concept demonstrated definite social, or at least experiential, potential, the group reached a unanimous vote for the most burning and generally applicable issue: first-year students and people new to UL are constantly overwhelmed by the maze of campus. It was decided that the corresponding app would directly cover the wants and needs of the user group, as well as include secondary aspects in the design: awareness of events, accessibility, and safety.

(Prototype images in annex 1.)



Mind Mapping: After completion of our brainstorming discussion, our team moved on to conduct a more organized mind-mapping task to structure and visually consolidate ideas related to the UL Campus Nav App idea. The mind map was centered around the “UL Campus Nav” concept and further conducted around the five main categories influenced the core of app’s design features, Design, Benefits, Users and Purpose.

The visual representation of the primary design areas also helped identify their relationships and establish a connection between the provided functionality, design solutions, and target users’ needs. Ultimately, this factor contributed to comprehending the overall experience of the limited users and how the application could ensure a memorable experience.

2.2 The Group Project Developed Features:

Several potential features are identified in the mind mapping session which were aimed at making navigation simple, efficient and informative. The features are as follows:

- **360° Map View:** ability for users to try on a virtual “walker” interface, see a student in casual clothes, and immerse themselves through the eyes of an observer in the thick of things. The most interesting part of the view is that this is a keen shot and zoomable in and out view. The user can install this into their VR by playing games and watching entertainment videos if connected.
- **Event information:** integrated ongoing and upcoming UL events.
- **Indoor Navigation:** leading a student through building interiors, dining space, lecture rooms, laboratories, libraries, and other similar facilities.
- **Facility Locator:** prohibits a guide to the cafeteria, stationary room, and ATM along with restaurants and other off campus facilities.
- **Opening hours information:** accessible and unavailable times for buildings and departments.

The group first phase of the project laid a very strong foundation for the UL Nav App Development, providing essential features and valuable insights into its functionality and user interaction. This groundwork serves as the starting point for the individual redesign phase, focussing on refining UL Nav App to better meet the student and the other user’s needs.

3. Testing:

3.1 Type of Evaluation Selected

Testing is a crucial part of the design process for ensuring that the UL Campus Navigation App effectively supports users in finding their way around the university. It allows designers to uncover usability issues, observe real user behaviour, and confirm whether the interface and features work as intended. Through hands-on interactions with the prototype, testing reveals where users struggle, what features they understand easily, and how well the app communicates essential navigation information. The purpose of this evaluation was to assess the usability of the UL Campus Navigation App, identify areas that caused confusion, and determine opportunities for improvement.

This included: Examining how users interpreted the interface, how easily they could search for a building, and how clearly, they understood the route preview and live navigation screens.

Feedback: Feedback gathered during testing highlighted several key issues: users found the interface visually confusing at first glance, many building names lacked clear explanations, and the navigation bar icons did not feel intuitive or meaningful. The step-by-step route preview also caused difficulty, as instructions like “Step 1: Head west (50 m)” or “Step 2: Enter building via east entrance (30 m)” were not always easy to follow. Accessibility options and map layers were also unclear, leading to uncertainty about which routes or entrances were suitable for individual needs.

By identifying these problems, the testing phase provided essential insights that directly informed the redesign. The evaluation guided improvements to interface clarity, visual hierarchy, accessibility indicators, and the overall navigation flow, from the home screen to live guidance. This ensured that the new design better aligns with users’ expectations and supports a more confident, stress-free wayfinding experience across the UL campus.

3.2 Testing Method:

A formative usability evaluation was conducted using a high-fidelity prototype of the UL Campus Navigation App group phase design. The aim was to observe how users interacted with the app in a realistic context and identify issues that hindered their ability to complete core navigation tasks.

Task based testing approach was used, where few participants students were asked to carry out specific actions such as searching for a building, viewing building information, checking the route preview, and following step-by-step directions. The sessions combined think-aloud protocol with direct observation, allowing users to express their thoughts, confusion, and expectations while interacting with the prototype.

3.3 Participant Profile:

A small group of participants was selected to represent typical users of the UL campus. This included:

- **First-year students** unfamiliar with the full layout of the campus
- **International students** who may struggle with initial orientation
- **Visitors or non-UL members** who have no prior knowledge of building locations
- **Returning students or staff** who are familiar with some areas but not all

Participants varied in age, digital literacy, and familiarity with navigation apps. This diversity helped capture a broad range of user experiences and highlighted issues that different types of users encountered when navigating the prototype.

The sample was intentionally kept small (typical for formative testing), as the goal was to reveal common usability problems rather than achieve statistical completeness.

3.4 Key Findings:

The testing revealed several consistent usability issues that affected the clarity and overall usability of the app:

- **Interface felt confusing at first glance**

Participants struggled with the initial screen, which presented many buildings without clear explanations. Users were unsure what each building represented or which one to select.

- **Poor discoverability of navigation bar features**

Buttons and icons in the navigation bar did not feel intuitive, and users frequently hesitated or tapped the wrong icon because the purpose was unclear.

- **Map and route preview were difficult to understand**

Step-by-step instructions such as “Step 1: Head west (50 m)” and “Step 2: Enter building via east entrance (30 m)” were not always easy to interpret. Users wanted clearer visual cues and simpler directions.

- **Cognitive overload during navigation tasks**

Participants felt overwhelmed by the amount of information presented on some screens, especially maps with multiple elements or unclear layers.

- **Uncertainty about map layers and visual elements**

Some users did not understand what different symbols or map modes represented, leading to confusion during route selection.

Overall, these findings confirmed the need for a clearer, more intuitive, and more supportive navigation experience.

3.5 Implications for Redesign

The insights gathered during testing directly shaped the next phase of the redesign. Key improvements included:

- **Simplifying the user interface**

A cleaner layout and stronger visual hierarchy were introduced to reduce confusion and help users understand the app immediately on launch.

- **Redesigning the navigation flow**

The app was restructured into a clear four-step journey **Home** → **Place Detail** → **Route Preview** → **Live Navigation**, making the experience more predictable and easier to follow.

- **Enhancing the step-by-step route preview**

Instructions were redesigned to be more visual and easier to interpret, with a mini-map, clearer icons, and simpler text.

- **Making accessibility visible and meaningful**

New indicators for accessible routes and entrances were added to help users with mobility needs make informed decisions

- **Improving icon clarity and navigation bar behaviour**

Icons were redesigned for clarity, and their functions were made more consistent across the app, improving user confidence.

4. Exploring Design Directions:

4.1 Identifying Possible Directions

Following the usability testing and analysis of user feedback from the group prototype, several potential directions emerged for the redesign of the UL Campus Navigation App. These directions focus on simplifying the interface, improving clarity, and supporting users as they move through the campus. The goal is to create a navigation experience that is intuitive, accessible, and specifically tailored to the complexity of the UL environment.

Based on the issues identified during testing—such as confusing map elements, unclear navigation bar actions, and difficulty understanding route previews—the following design directions were identified as the most valuable for further development.

- **Improve Interface Clarity and Reduce Visual Overload**

Users found the original layout visually overwhelming, particularly the home screen and map views. A key direction is to simplify the visual hierarchy, reduce unnecessary elements, and present information in smaller, more digestible sections so users can understand the app briefly.

- **Enhance Discoverability of Core Features**

Participants struggled to understand what certain icons or buttons meant. Improving feature discoverability includes refining the navigation bar, adding clearer labels, and making actions more predictable so users can easily find what they need without guessing.

- **Redesign the Route Preview for Clarity**

Users found step-by-step directions such as “Step 1: Head west (50m)” difficult to interpret. A direction for redesign is to make the route preview more visual, more

intuitive, and more consistent using clearer icons, a simplified mini-map, and shorter instructions to support quick understanding.

- **Strengthen Accessibility and Inclusive Navigation**

Users highlighted uncertainty around accessibility routes and entrances. A dedicated design direction is to integrate accessibility information directly into the map and route preview, ensuring that users with mobility needs can easily find suitable pathways.

- **Simplify Navigation Flow from Start to Finish**

The original multi-step flow felt disjointed and unintuitive. A central design direction is to establish a clean four-stage flow:

Home - Building Detail - Route Preview - Live Navigation,

ensuring each step clearly leads to the next without confusion

4.2 Brainstorming and Mind Mapping

To explore the redesign directions in more depth, a brainstorming and mind-mapping process was carried out. This phase helped generate a wide range of ideas, visualise potential solutions, and organise concepts based on the usability problems identified during testing. By mapping out thoughts freely and without constraints, it became easier to identify patterns, opportunities, and the most promising design directions for the UL Campus Navigation App.

Brainstorming Process

Brainstorming focused on addressing the main issues users experienced, such as confusing map elements, unclear navigation flows, and low feature discoverability. Ideas were written down rapidly without filtering, allowing creativity to guide potential improvements. Key brainstorming themes included:

- How to make buildings easier to recognise
- Ways to simplify the navigation bar
- How to make the route preview more readable
- Better ways to present step-by-step directions
- Stronger accessibility indicators
- Reducing visual overload on the map
- Creating a clearer start-to-end journey

This approach generated multiple design possibilities that could later be refined during the prototyping stages.



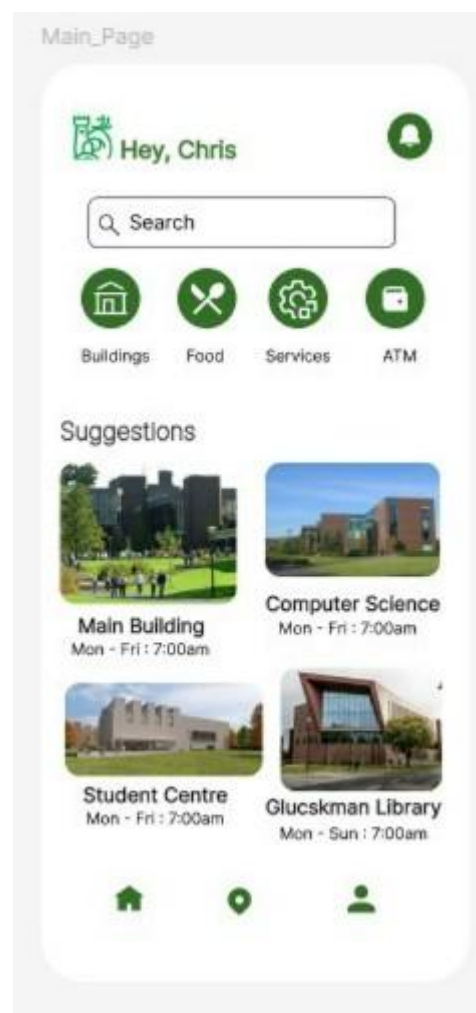
Brainstorming Image

Mind Mapping

A mind map was created to visually organise the brainstorming ideas and connect related concepts. The central node of the mind map was **“UL Campus Navigation Redesign”**, and branches expanded into key areas influenced by user feedback:

1. Home Screen

- Simplified layout
- Clear search bar
- Highlight common locations
- Include campus categories (cafés, lecture halls, services)



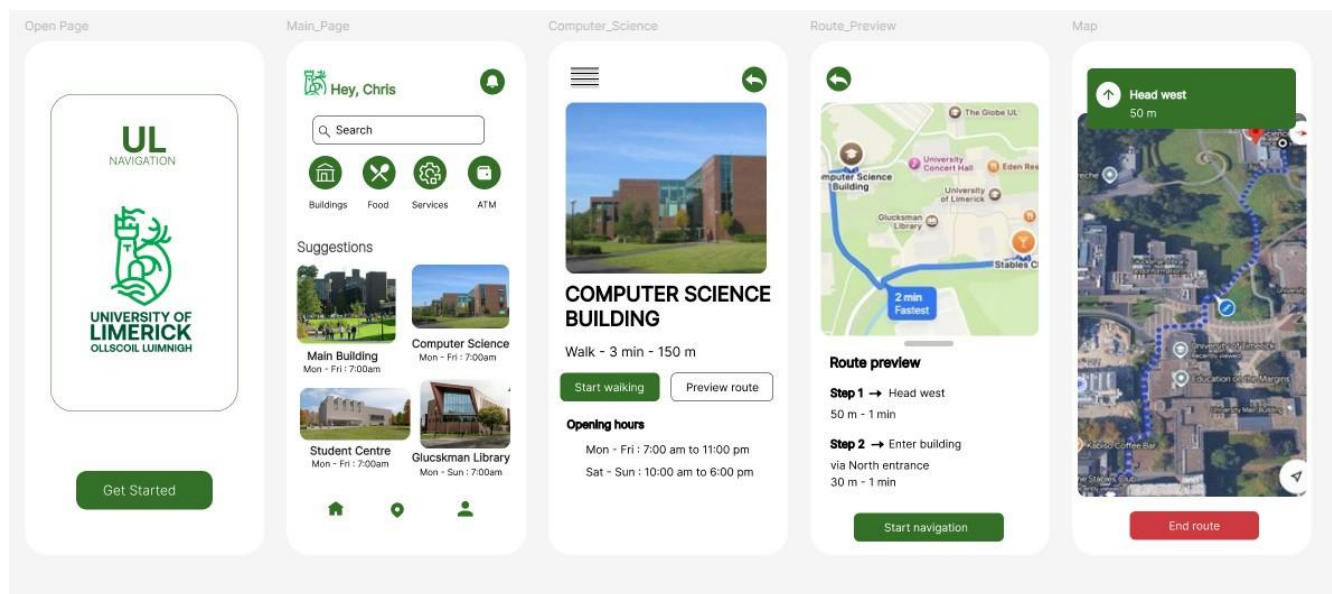
Home Screen:

2. Building Information

- Add images and short descriptions
- Include accessibility details
- Display opening hours or key features
- Show entrances on the mini map

3. Navigation Flow

- Consistent 4-step structure (Home - Building - Route - Navigation)
- Clear transitions between screens
- Predictable button placement



New Redesign Nav App

4.Route Preview

- Visual step list
- Short, action-based instructions (e.g., "Head west 50 m")
- Simplified mini map with a highlighted route

-
- Accessible route options

5. Navigation Bar

- Replace unclear icons
- Add text labels under icons
- Prioritise the most important actions
- Create a more intuitive layout



Old Design:



New Design:

Outcome of the Brainstorming & Mind Mapping

The brainstorming sessions and mind map helped clarify the redesign focus. These activities made it easier to prioritise features, structure the app's flow.

This exploration laid the foundation for the redesigned interface, ensuring that the final prototype directly addresses user needs while aligning with principles of clarity, accessibility, and discoverability.

5. Designing:

5.1 Features

The app-based redesign of the UL Campus Navigation App focuses on improving clarity, enhancing usability, and creating a smooth, confidence-building wayfinding experience for students, staff, and visitors. The following features were developed based on user feedback, usability testing results, and the need to simplify navigation across a large and complex campus environment.

1. Simplified Home Screen

A cleaner, more organised home screen reduces cognitive load and helps users understand the app instantly. The layout features a prominent search bar, clear categories (**Buildings, Services, Cafés, ATM**), and quick access to popular locations. This ensures users can begin their journey without confusion.

2. Improved Building Information Pages

Each building now includes:

- A photo for quick visual recognition and
- A short description
- Walking time and distance
- Key points of interest inside the building
- This gives users the context they previously lacked, making destinations easier to identify and select.

3. Clear Route Preview

The redesigned route preview provides simple, step-by-step instructions such as:

- Step 1: Head west (50 m)
- Step 2: Enter the building via the north entrance (30 m)

A small mini map illustrates the route visually, reducing confusion and helping users understand their direction before starting navigation.

4. Refined Navigation Bar

The new navigation bar replaces unclear icons with simpler visuals and text labels. This improves feature discoverability and helps users understand what each button does without guessing. Only essential functions are included to avoid unnecessary clutter.

5. Consistent 4-Step Navigation Flow

To improve the user journey, the app now follows a predictable and intuitive flow:

Home - Building Info - Route Preview - Live Navigation

This structure helps users understand where they are in the process and what comes next.

5.2 Design Rationale

The design decisions for the UL Campus Navigation App were driven by the issues discovered during usability testing and guided by principles of clarity, discoverability, and accessibility. The overall goal of the redesign was to reduce confusion, help users recognise buildings more easily, and support a smoother navigation flow across the campus.

A key rationale was simplifying the interface. Users previously described the layout as “crowded” and “unclear,” especially when viewing multiple buildings without labels or meaningful context. To address this, visual hierarchy was strengthened through larger headings, cleaner spacing, and improved building cards. Icons were paired with labels to reduce ambiguity and make essential actions more intuitive.

Another major motivator for the redesign was the difficulty users experienced in understanding map information. Many could not decode the symbols or interpret instructions such as “Head west 50 m.” The new design shifts toward more visual guidance using clearer route lines, building photos, directional arrows, and simplified step instructions. This reduces reliance on text and supports quicker comprehension.

Accessibility was also central to the rationale. The original prototype lacked clear indicators of accessible routes or entrances, which created uncertainty for users with mobility needs. The redesign incorporates accessibility icons, easy-to-toggle accessible route options, and descriptive entry information.

Finally, the rationale focused on improving the end-to-end journey. The redesign establishes a consistent four stage flow **Home - Building - Route Preview - Navigation** ensuring users always know where they are and what comes next. This structure reduces cognitive load and supports predictable, user-friendly interactions.

5.3 Wireframes

To translate early design ideas into a structured layout, low-fidelity wireframes were created for each major screen in the navigation flow. The purpose of these wireframes was to determine layout, hierarchy, and interaction before adding final visual style.

Home Screen Wireframe

- Search bar placed prominently at the top
- Category buttons (Buildings, Services, Cafés, ATM)
- A simplified list of recommended or frequently searched places
- Clear three-tab navigation bar at the bottom

Building Information Wireframe

- Large building image for quick recognition
- Title, distance, and opening and closing time
- Accessibility icon and brief building description
- CTA buttons: Start Walking and info

Route Preview Wireframe

- Mini map displaying the highlighted route
- Step list in simple numbered format
- Step 1: Head west (50 m)

-
- Step 2: Enter via north entrance (30 m)
 - Accessibility toggle for alternative routes
 - CTA button to begin live navigation

Live Navigation Wireframe

- Full map view with a clear route path
- Step bubble at the bottom with the current direction
- Top bar showing ETA, distance remaining, and step number
- “End Route” button anchored at the bottom

These wireframes provided the foundation for iterating on layout and functionality, allowing quick testing and refinement before final high-fidelity prototyping.

5.4 Final Prototype Screens

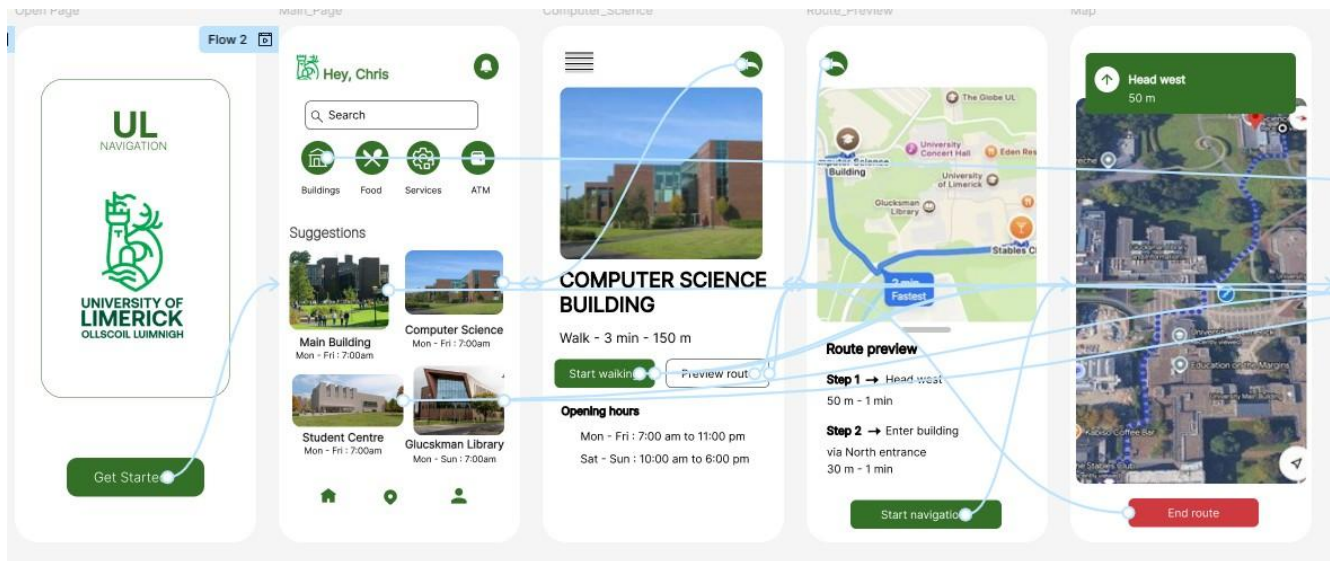
The final prototype transforms the wireframes into polished, high-fidelity screens that reflect the full visual direction of the redesigned UL Campus Navigation App. Each screen is designed for clarity, accessibility, and ease of use, ensuring users feel confident when navigating the campus.

1. Home Screen

The completed home screen presents a clean and welcoming layout with a clear search bar, recognisable category icons, and easy access to commonly used destinations. Visual clutter has been removed, improving first-glance understanding.

2. Building Information Screen

The final design features a large building photo and clear labels. Key details such as walking distance, time, and accessibility are prominently displayed. The Start Walking and Preview Route buttons are visually distinct and easy to find.



Final Prototype:

3. Route Preview Screen

This screen now includes a simplified mini map with a highlighted polyline route. Step instructions are concise, numbered, and supported by icons. The layout allows users to quickly assess the route before starting navigation. Accessibility routes are clearly marked.

4. Live Navigation Screen

The navigation experience now feels more intuitive and familiar, like modern map apps. The map is clean, with bold route lines and a visible position indicator. ETA, distance, and step number are clearly displayed at the top. The bottom step card provides short, actionable instructions such as “Turn right in 20 m.”

The prototype successfully addresses user pain points by improving clarity, reducing cognitive load, and guiding users through a predictable, well-structured navigation journey.

6. Evaluation of the Redesign

The redesign of the UL Campus Navigation App was evaluated to determine whether the proposed improvements successfully addressed the usability issues identified during the initial testing phase. This evaluation focused on clarity, discoverability, navigation flow, accessibility, and overall user confidence when interacting with the app. The goal was to understand how the redesign impacted the user experience and to assess the effectiveness of the new interface and interaction model.

6.1 Improved Interface Clarity

Users responded positively to the simplified layout and cleaner visual hierarchy. The reduced visual clutter made the home screen and map views easier to understand briefly. Clearer building cards, larger text labels, and consistent iconography significantly improved first-time comprehensibility. Participants noted that they no longer felt “**overwhelmed**” or “**lost**” when opening the app.

6.2 Enhanced Feature Discoverability

One of the most successful outcomes of the redesign was the improved discoverability of core features. Adding text labels to navigation bar icons, refining the placement of key buttons, and creating a predictable four-step flow helped users understand how to move through the app. Participants found it easier to access building information, view route previews, and begin navigation without hesitation.

6.3 More Understandable Route Instructions

The redesigned route preview with short, clear steps such as “**Head west (50 m)**” and “**Enter via north entrance (30 m)**” was far easier for users to interpret. Visual cues such as arrows, simplified icons, and a cleaner mini map helped reduce cognitive load. Users expressed greater confidence in understanding the directions before starting navigation.

6.4 More Predictable and Intuitive Navigation Flow

The consistent flow **Home - Building Info - Route Preview - Live Navigation** helped users understand where they were in the process. The new layout reduced confusion and created a smoother transition between screens. Participants described the experience as “more organised” and “easy to follow.”

6.5 Increased User Confidence and Reduced Cognitive Load

One of the main goals of the redesign was to reduce the mental effort required to navigate the campus. The improved interface, clearer instructions, and consistent flow collectively resulted in greater user confidence. Participants no longer felt unsure about what buttons did, where to look next, or how to interpret the route.

6.6 Remaining Opportunities for Improvement

Although the redesign addressed many key issues, a few opportunities for further enhancement were identified:

- Some users wanted an optional **voice navigation** feature for hands-free guidance.
- A few participants requested **landmark-based directions** (e.g., “Turn right after the footbridge”).
- Some users suggested adding a **dark mode** for low-light environments.

These suggestions will be considered for future iterations of the design.

Overall Evaluation

Overall, the redesign successfully resolved most of the usability issues highlighted during the group testing phase. Users found the new interface clearer, the route preview more understandable, and the overall navigation flow more intuitive. The improvements greatly reduced confusion and improved user confidence when navigating the UL campus. The redesign demonstrates how user-centred design, iterative testing, and clearer interaction patterns can transform a complex navigation system into a more effective and accessible tool for the UL community.

7. Conclusion:

The redesign of the UL Campus Navigation App represents a significant step toward creating a clearer, more intuitive, and user-centred wayfinding experience for students, staff, and visitors at the University of Limerick. Through a structured process grounded in user-centred design principles, the project analysed real user challenges, translated insights into meaningful design directions, and developed a refined prototype that directly responds to those needs.

Usability testing from the group phase highlighted crucial issues such as confusing map elements, unclear navigation bar icons, poor route comprehension, and limited accessibility support. These findings formed the foundation of this redesign, informing decisions aimed at reducing cognitive load and improving clarity at every stage of the navigation journey.

The final prototype demonstrates substantial improvements. A simplified home screen, clearer building information pages, more understandable route previews, and a streamlined four-step navigation flow work together to create a predictable and confidence building experience. Enhanced accessibility features and a cleaner visual design further support inclusivity and ease of use. Overall, the redesign not only improves the functional aspects of navigation but also considers the emotional experience of users who may feel overwhelmed when navigating a large and unfamiliar campus.

This redesign shows the value of iterative testing, thoughtful analysis, and continuous refinement. While the prototype addresses many core usability challenges, future opportunities such as voice navigation, landmark-based directions, and dark mode provide room for continued enhancement.

The UL Campus Navigation App redesign therefore stands as a meaningful evolution of the original group concept, transforming it into a stronger, more coherent, and more user-friendly solution. This project underscores the importance of designing with users in mind and demonstrates how effective interaction design can directly improve the everyday experiences of the UL community.

8.Reference:

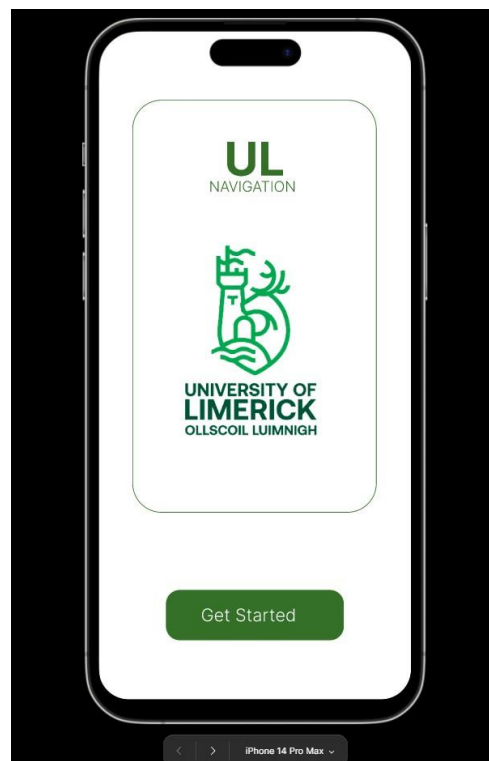
Norman, D. A. (2014). *The Design of Everyday Things* (Revised and Expanded Edition). New York: Basic Books.

University of Limerick (2024). *Campus Map and Facilities Overview*. Retrieved from <https://www.ul.ie/>

Figma (2025). *Figma: Collaborative Interface Design Tool*. Retrieved from <https://www.figma.com>.

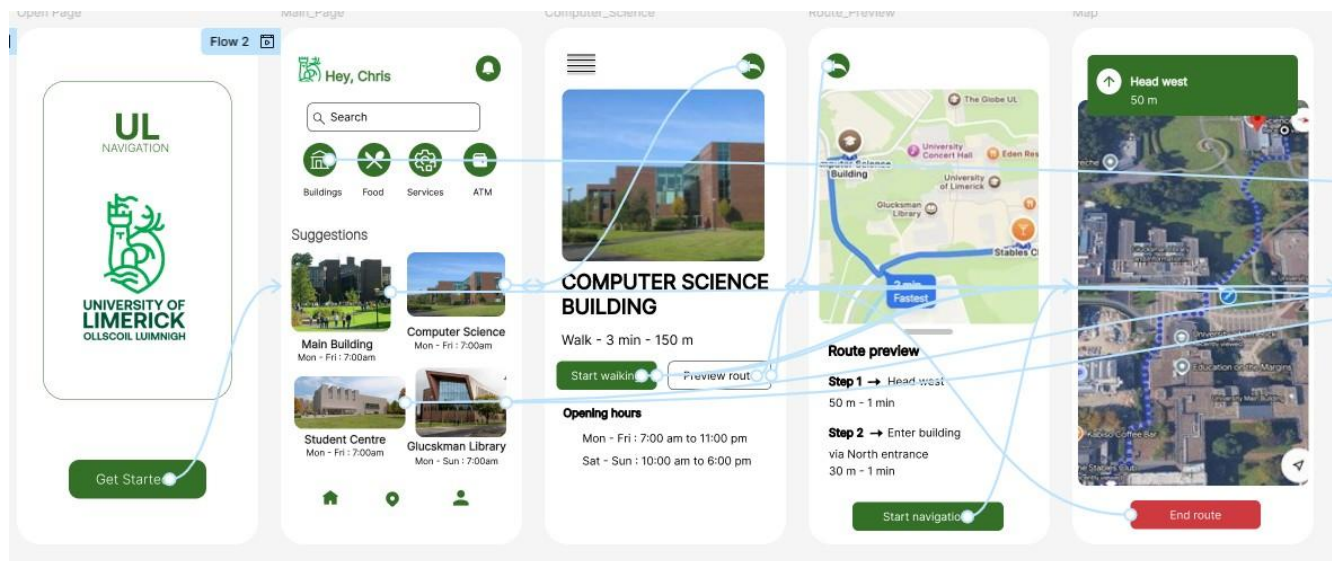
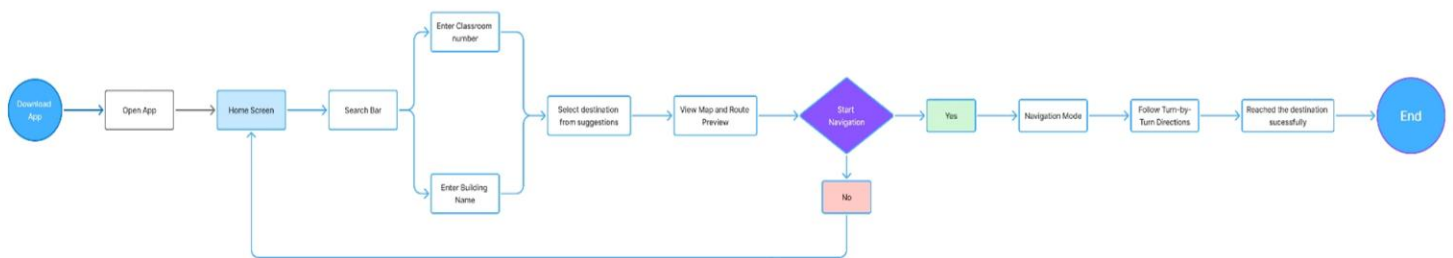
Figma Link for the final Prototype:

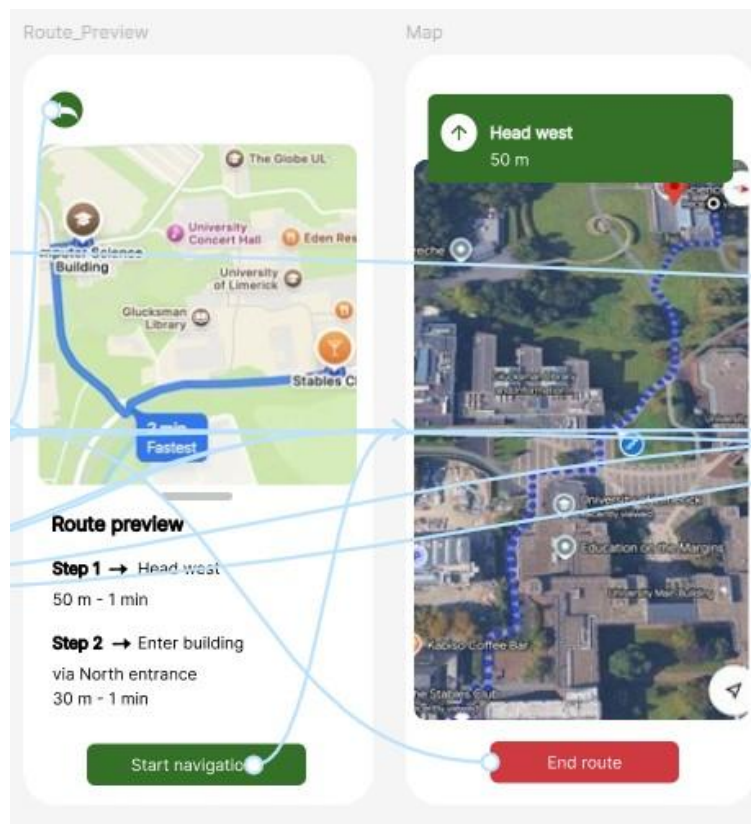
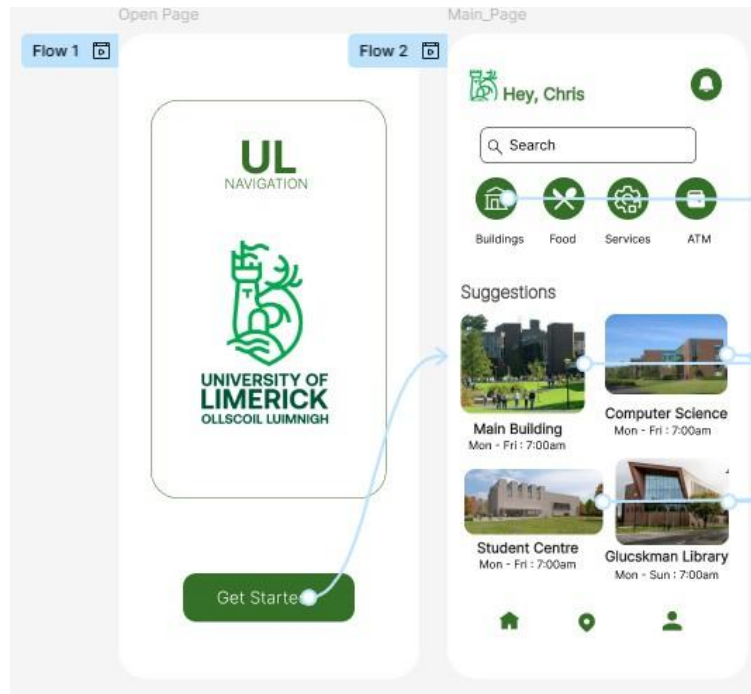
<https://www.figma.com/proto/LigkfW0zKSaaU5zdad63iX/Untitled?page-id=1%3A2&node-id=2-8&viewport=-393%2C484%2C0.25&t=Gp8NiBRelE049N8z-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=2%3A8&show-proto-sidebar=1>

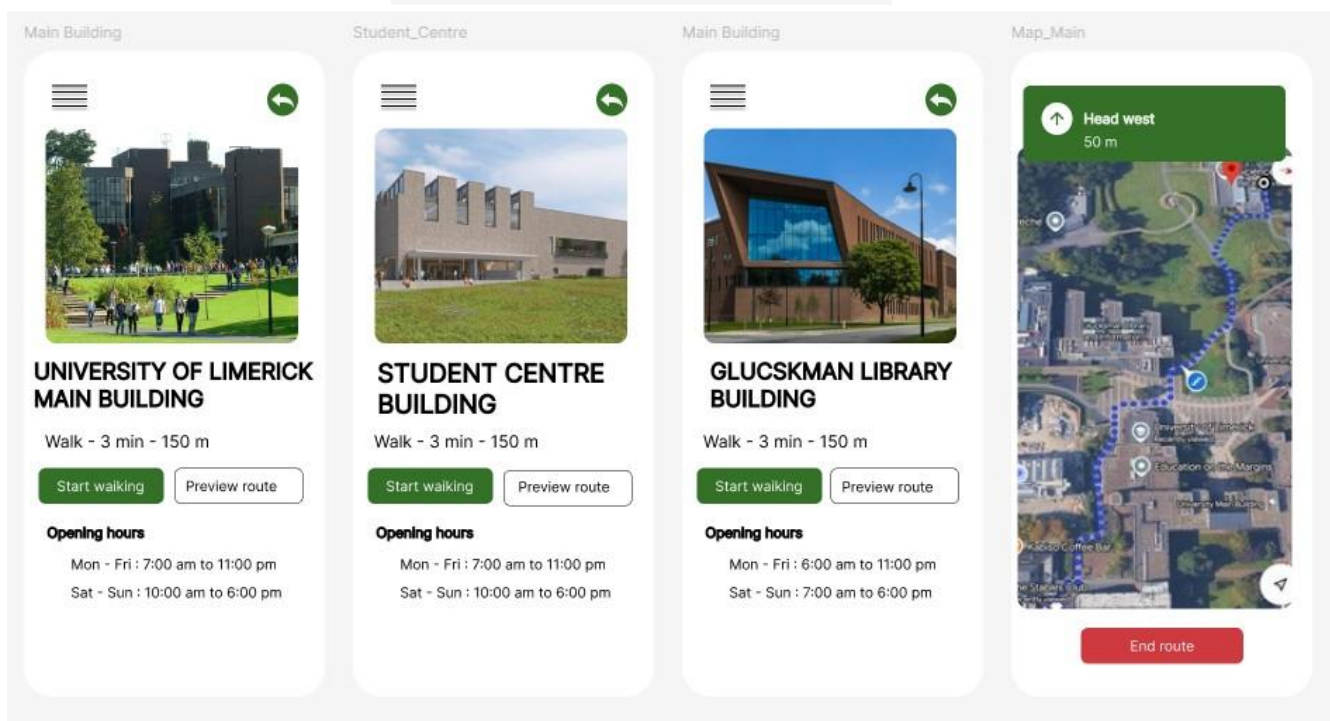
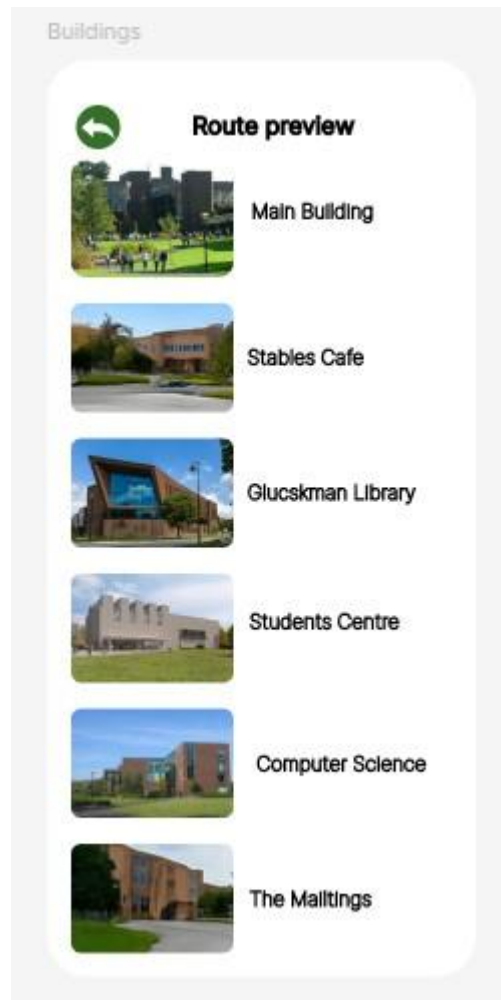


9. Annex 1:

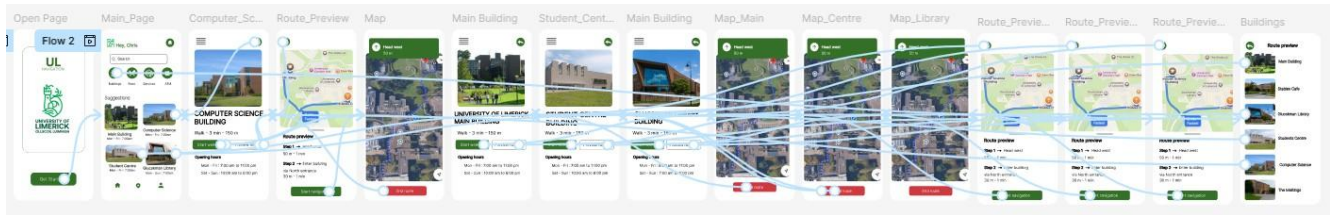
User Flow of UL Campus Nav App:







Prototype Connection:



THANKYOU